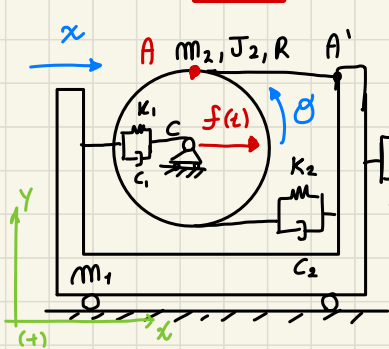




ESE 3 es 1



WRITING eq of motion for a MULTI-DOF MECHANICAL SYSTEM (KINEMATICALLY LINEAR here)

linear eq of motion!

- 2 RIG bodies
- C-shape slider
- DISK constrained by cart
- DISK ROTATES + MOVES HORIZ
- + WIRE connecting CART to disk
- + SPRING/DAMPER

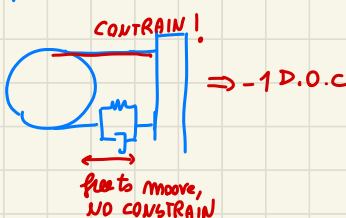
DOF? (2 DOF) $\downarrow$ in fact

2 R.B
SLIDER
CART
WIRE

6 DOF
- 2 D.O.C
- 1 D.O.C
- 1 D.O.C

$\downarrow$
2 DOF remaining

the WIRE has 2 BRANCH



INDEPENDENT COORD

x horizontal movement of Body 2 (slider)
(give all body 1 motion, nothing about body 2 $\rightarrow$ DISK)

DISK motion $\Rightarrow$ using x_c as HORIZ movement of C
OR take DISK ROT θ !

IN FACT

$\left. \begin{array}{l} \dot{x} = \text{const} \\ \dot{\theta} = \text{const} \end{array} \right\} \begin{array}{l} \text{ALL} \\ \text{FROZEN!} \end{array}$

$$\underline{x}(t) = \begin{pmatrix} x \\ \theta \end{pmatrix}$$

WRITE eq of motion... LAGRANGE (need T, V, D, SWR)
(VECTOR FORM!) $\rightarrow$ involving $x, \dot{x}, \ddot{x}, \theta, \dot{\theta}, \ddot{\theta}$

found the MATRIX use
know how to

WRITE the final eq of

motion as the STANDARD eq $(m)\ddot{x} + (c)\dot{x} + (k)x = f(t)$

SCALAR FORM

$\leftarrow$ involving square symm positive matrix $[m], [c], [k]$
multiplied by our independent coordinates

$$T = \frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 \dot{x}_2^2 + \frac{1}{2} J_2 \dot{\omega}_2^2$$

B1 can only
translate
along x
(NO y or $\dot{\theta}$)

C moves
along x
C = G uniform mass disk

entire system
kin energy

$\hookrightarrow$

Vector approach

$$\dot{\underline{y}}_m = \begin{pmatrix} v_{G1x} \\ v_{G2x} \\ \omega_2 \end{pmatrix}$$

$$[m_{ph}] = \text{diag}\{m_1; m_2; J_2\}$$

↓ Jacobian matrix to relate $\dot{\underline{y}}_m$ to $\dot{\underline{x}}$

$$\dot{\underline{y}}_m = [\lambda_m] \dot{\underline{x}} \quad \text{for sure } [\lambda_m] \text{ is a MATRIX } (3 \times 2)$$

build using the

R.B kinematic on this example...

$$[\lambda_m] = v_{G1x} \begin{bmatrix} \dot{x} & \dot{\theta} \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}$$

Fill in the MATRIX using

SUPER-IMPOSITION principle, Because

Linear system

(OR if NONLIN we linearize all syst

↪ replace with linearize eq. around equilibrium)

use assumption linearity of SYSTEM when WRITING the Jacobian!

● total v_{G1x} will be super imposition of two effects,

↓ $\dot{x} \neq 0, \dot{\theta} = 0$ (freeze disk) + $\dot{x} = 0, \dot{\theta} \neq 0$ (freeze slider)

With this approach you fill the matrix easily considering 2 single system 2DOF

freeze ROTATION... $\dot{x} \neq 0, \dot{\theta} = 0$

⇒ TO FILL FIRST COLUMN of λ_m ($\dot{x}$ EFFECT)

TRANSLATORY ←

= all points moves

at the same way

DISK JUST TRANSLATORY MOTION

$v_{G2x} = v_2$ any point of DISK velocity!

you can choose a practical point

$\dot{x} \neq 0 \rightarrow B1$ move fully horizontally TRANSLATING by $\dot{x}$

(all points move at same way) $v_{Gx} = 1 \dot{x}$ $\lambda_{m,1,1}$ element

for DISK analysis of

motion we took (A) ⇒

DISK translate BUT connected by

UN-STRETCHABLE WIRE to the slider

$v_{G2x} = v_{A'} = \dot{x}$ ← AA' constant length if $\dot{\theta} = 0$

same for point C → $v_{G2x} = v_C = \dot{x}$

● frozen → NO ROTATION... $\omega_2 = 0$

• IF FROZEN SLIDER $\dot{\theta} \neq 0, \dot{x} = 0$ (second column)

IT IS like clamping

the slider to the floor

⇒ $v_{G1x} = 0$ STILL/FROZEN

While the DISK ROTATE (NOT TRANSLATORY MOTION)

DISK have different velocity $\forall$ point (translatory, vertical / horiz..) NOT well by

RESPEC TRANSLATIO  all same velocity

IS ROTOTRSL.



different vel $\leftarrow$

ROTATING
DISK like on
our case...

$$V_c \neq V_A$$

IN ROTATION

easy way to solve the problem

is FIND the (C.I.R) and use one point P with $V_P = 0$

HERE the DISK has 0 velocity

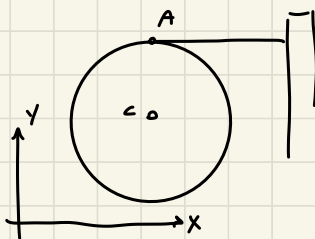
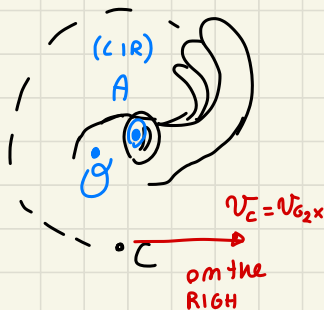
on point (A) because $V_A = V_{A'} = 0$ (WIRE)

$A \equiv$ CIR of DISK when $\dot{x} = 0$

entire DISK ROTATE around A (NOT around the center!)

Found CIR every velocity is $\perp$ to line connecting point with CIR
with magnitude $\{ \dot{\theta} \text{ (angular ROT)} \} \cdot \{ \text{distance} \}$

$$|\vec{V}_c| = R \dot{\theta} \text{ with direction respect RIGHT HAND}$$



$$V_{G2x} = + R \dot{\theta}$$

and clearly $\omega_2 = 1 \dot{\theta}$

we totally fell $[\lambda_m]!$

FOR ANY Nulls DOF... once found $[\lambda_m]$



$$[\lambda_m] = \begin{bmatrix} 1 & 0 \\ 1 & R \\ 0 & 1 \end{bmatrix}$$

(MASS MATRIX)

$$T = \frac{1}{2} \dot{\vec{x}}^T [\underline{m}] \dot{\vec{x}}$$

square matrix
symmetric
of order 2

$$[\underline{m}] = [\lambda_m]^T [m_{ph}] [\lambda_m]$$

(on WRITTEN exam we
do it with MATLAB!)

$$[\underline{m}] = \begin{bmatrix} m_1 + m_2 & m_2 R \\ m_2 R & m_2 R^2 + J_2 \end{bmatrix}$$

SYMM MATRIX 2x2

• Potential energy $V = V_{el} + V_g$ unusual

HERE it easy to see G_1, G_2 cannot move vertically $V_g = \text{constant}$ and for slime, Because system kinematic is linear

$\Rightarrow$ effect of pre-load cancel with each other, because linear (K_1, K_2, K_3) part of quadratic expression, on V_{el} we

only simple case where we deal only with the Δl from equilibrium

With sign convention $\leftarrow \text{---} \text{---} \text{---} \rightarrow$ (+) positive of elongate

$$V = \frac{1}{2} K_1 \Delta l_1^2 + \frac{1}{2} K_2 \Delta l_2^2 + \frac{1}{2} K_3 \Delta l_3^2 = \frac{1}{2} \underline{y}_K^T [K_{ph}] \underline{y}_K$$

$\downarrow$ IN VECTOR FORM:

(LINEAR SYSTEM)

LINEAR

Relationship

through Jacobian

$$\underline{y}_K = \begin{pmatrix} \Delta l_1 \\ \Delta l_2 \\ \Delta l_3 \end{pmatrix}$$

$$[K_{ph}] = \text{diag} \{ K_1; K_2; K_3 \}$$

$$\underline{y}_K = [\lambda_K] \underline{x}$$

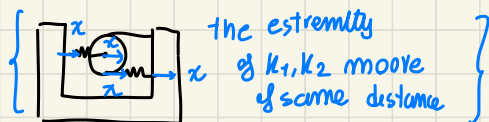


super-imposition

• $x \neq 0, \theta = 0$ all disk point has same displacement (TRANSLATE)

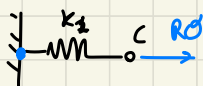
$\Delta l_1 = 0 \quad \Delta l_2 = 0$ DISK FREEZE on the slider

$\Delta l_3 = -x$ compressing spring 3 with $x > 0$

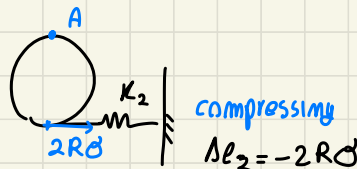


• $x = 0, \theta \neq 0$ slider fixed, clear that $\Delta l_3 = 0$

While, because C.I.R is point A, When ROTATE of θ



$$\Delta l_1 = R\theta$$



compressing $\Delta l_2 = -2R\theta$

so found $[\lambda_k] \dots$ $V = \frac{1}{2} \underline{\dot{x}}^T [K] \underline{\dot{x}}$

$$[K] = [\lambda_k]^T [K_{ph}] [\lambda_k] = \begin{bmatrix} K_3 & 0 \\ 0 & R^2(K_1 + 4K_2) \end{bmatrix}$$

this is the IORD elastic

stiffness matrix $\rightarrow$ here II ORD and GRAVIT

ONE because we have a

KINEMATICALLY linear system

(so linear kin. relation)

• DISSIPATION

(damping matrix)

$$D = \frac{1}{2} \underline{\dot{y}}_k^T [C_{ph}] \underline{\dot{y}}_k$$

because SPRING// DUMP

$\forall$ spring we have parallel damper

$$\rightarrow [C_{ph}] = \text{diag} \{ C_1 ; C_2 ; C_3 \}$$

because each dump // spring

$\downarrow$

the Jacobian $\underline{\dot{y}}_k = [\lambda_k] \underline{\dot{x}}$ it is the same $[\lambda]$ as SPRING

$$[C] = [\lambda_k]^T [C_{ph}] [\lambda_k] = \begin{bmatrix} C_3 & 0 \\ 0 & R^2(C_1 + 4C_2) \end{bmatrix}$$

Just simply by replacing
 K_i with C_i

• VIRTUAL WORK of t dependent forces

$$\delta W_{nc} = \delta x_c \cdot f(t) = \delta y_F^T \underline{f}_{ph}$$

even if it seems useless

complication, it is consistent
with MATRIX approach

$$\delta y_F = [\delta x_c] \rightarrow \text{JACOBIAN MATRIX}$$

$$\underline{f}_{ph} = [f(t)] \quad \delta y_F = [\lambda_F] \delta \underline{x}$$

$\downarrow$
SHORTCUT to
write $\lambda_F \dots$

$$[\lambda_F] = \begin{bmatrix} \delta x & \delta \theta \\ 1 & +R \end{bmatrix} \delta x_c$$

because $\dot{x} = \frac{\partial x}{\partial t}$ velocity is just a ratio of infinitesimal vector over a
scalar quantity (dt)

$\downarrow$
so the

relationship between δx_c and δx , $\delta \theta$ is exactly same

$\downarrow$ relationship between $\dot{x}_c$ and $\dot{x}$, $\dot{\theta}$! so

this is the second row of $[\lambda_m]$ expressing $\dot{v}_{cx} = \dot{x}_c$

$$\Rightarrow \delta W_{nc} = \delta \underline{x}^T \underbrace{\underline{f}(t)}$$

$$\underline{f}(t) = [\lambda_F]^T \underline{f}_{ph} = \begin{pmatrix} f(t) \\ R f(t) \end{pmatrix}$$

so we can WRITE the eq of motion as

$$[m] \ddot{\underline{x}} + [c] \dot{\underline{x}} + [k] \underline{x} = \underline{f}(t) \quad \text{Equation of 2 DOF system}$$

• SIMPLE system...

→ BUT for more DOF The finding of $\omega_i, \underline{x}^{(i)}$ and FRF of this... IS NOT EASY!
almost IMPOSSIBLE in close form

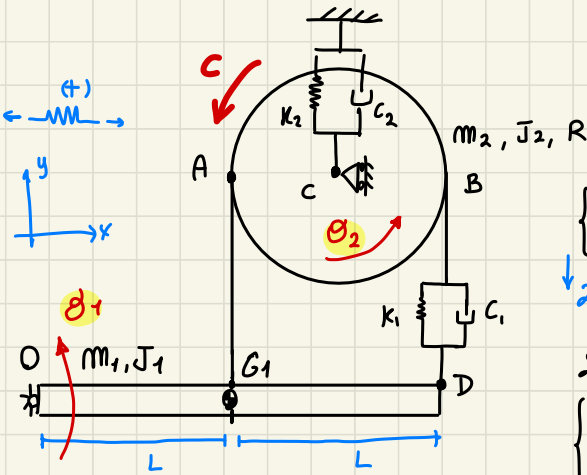


We use Numerical Values of the parameters $c, m, k, \dots$ and solve it in MATLAB



solve with MATLAB's function numerically

es2



dealing with 2 Rigid Bodies

• RIGID BAR in horizontal position as equilibrium

• DISK with a WIRE connected with BAR

$\left\{ \begin{array}{l} K_1, C_1 \text{ keep system in} \\ \text{vertical position} \end{array} \right\}$

2 DOF SYSTEM...

2 R.B $\rightarrow$ 6 DOF

Hinge O -2 DOC $\Rightarrow$ 2 DOF

Cont C -1 DOC

Wire fixed G -1 DOC

(While BD can change length)

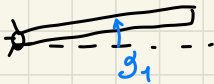
$$\underline{x} = \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix}$$

We take indep coord

- θ_1 (Rotation of BAR) (INDEPENDENT coord to choose!)
- vertical displacement y_c OR you can use DISK ROTATION θ_2

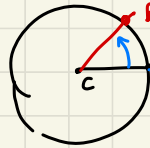
CAREFULLY $\rightarrow$ dealing with 2 ROTATIONS...

θ_1 for sure is absolute ROTATION of BAR



While Disk has Rotation respect

POLAR SYMMETRY Respect a Reference

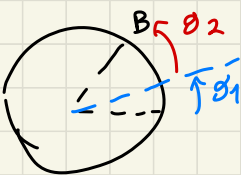


I have 2 choices!

Best is to measure θ_2 from the x axis of REF like do me in Body 1!

(θ_2 is absolute DISK ROT. Respect fixed Ref)

Other possible choice is to measure θ_2 from Bar direction



Respect Reference direction of the BAR

$\rightarrow$ RELATIVE ROTATION θ_2^R of DISK respect BAR

IN some cases you may have doubt to

use θ_2 or θ_2^R usually θ_2 is SAFER!

Unless you have advantage in using $\theta_2^R \rightarrow$ usually θ_2 BETTER

IF choosing θ_2^R carefully IT IS

$\leftarrow$ this implicate on the solution development

RELATIVE! $\Rightarrow$ SO With 2 ROTATIONS he carefully IF each θ IS RELATIVE OR ABSOLUTE

$\rightarrow$ here we decide θ_2 ABSOLUTE

WRITING of energetic quantities

• KINETIC ENERGY:

$$T = \frac{1}{2} \dot{\underline{y}}_m^T [m_{ph}] \dot{\underline{y}}_m \quad \dot{\underline{y}}_m = \begin{Bmatrix} V_{G1y} \\ \omega_1 \\ V_{G2y} \\ \omega_2 \end{Bmatrix}$$

$[m_{ph}] = \text{diag} \{m_1; J_1; m_2; J_2\}$
 $\uparrow$
 terms correspond to other $\dot{\underline{y}}_m$ terms
 scalar component of VECOCITY of ROTATION or center of mass

• Carry about small motion

around equilibrium, so about horizontal BAR equilibrium

• G_1 MOVES only vertically with APPROX

• ROTATING BAR

while DISK only ROTATE and moves vertically

and we have the JACOBBIAN

even IF NON LIN SYSTEM,

we analyze with super-imposit.

• $\dot{\theta}_1 \neq 0, \dot{\theta}_2 = 0$

WE CARE about small motion NEAR equilibrium $\rightarrow$ LINEAR.

When Bar Rotate of $\dot{\theta}_1$ point O

$\rightarrow$ fixed (CIR) $\rightarrow V_{G1y} = L \dot{\theta}_1$ clearly moving along $y > 0$
 $\omega_1 = 1 \dot{\theta}_1$

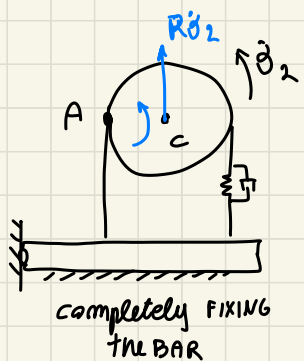
$$[\lambda_m] = \begin{bmatrix} L & 1 & 0 \\ -\frac{L}{1} & -1 & 0 \\ -\frac{L}{1} & 1 & R \\ 0 & 1 & 1 \end{bmatrix} \begin{matrix} V_{G1y} \\ \omega_1 \\ V_{G2y} \\ \omega_2 \end{matrix}$$

While IF $\dot{\theta}_2 = 0$ frozen DISK $\rightarrow$ JUST translating UP/DOWN with

same velocity of point $G_1 \rightarrow$ A same velocity of the WIRE connected to the BAR (so same terms related to $\dot{\theta}_1 \rightarrow L$)

While $\omega_2 = 0$

• When $\dot{\theta}_1 = 0$, $\dot{\theta}_2 \neq 0$



so Body 1 fixed $V_{G1y} = 0$, $\omega_1 = 0$
 the CIR of disk is A ($V_A = 0$)
 so point C moves as $+R\dot{\theta}_2 = V_{G2y}$
 and $\omega_2 = \dot{\theta}_2 \rightarrow +1$

once written the JACOBBIAN...
 as before

$$[m] = [\lambda_m]^T [m_{ph}] [\lambda_m] = \begin{bmatrix} (m_1 + m_2)L^2 + J_1 & m_2LR \\ m_2LR & m_2R^2 + J_2 \end{bmatrix}$$

NEXT we deal
 with POTENTIAL en.

$$V = V_{el} + V_g$$

here kinematic relationship ARE NOT
 LINEAR In fact G_1, D depends with
 $\sin(\theta_1)$ AND because $\sin$ function are
 such $\frac{\partial^2 \sin(\theta)}{\partial \theta^2} \Big|_{\theta=0} = 0$

special equilibrium
 position such that simplify solution
 WE expect some HESSIAN involved with
 heigh cam ... as $\mathbb{Q}$ because of NON LIN type



While once
 we have
 moment
 of G_1 respect
 this the
 moment of C, B
 are perfectly linear with $\theta_2 \Rightarrow R\theta_2, 2R\theta_2 \Rightarrow$ so Hessian $\mathbb{Q}$ also
 RESPECT θ_2

↓ this means Luckily the PRE-LOADS
 WEIGHT FORCES

on V_{el} we care
 only about Δe

(*)

cancel, provide
 $\theta_1, \theta_2 @ e_q$
 as $\mathbb{Q}$ effect

$$V = \frac{1}{2} \underline{y}_K^T [K_{ph}] \underline{y}_K \quad \text{with} \quad \underline{y}_K = \begin{pmatrix} \Delta l_1 \\ \Delta l_2 \end{pmatrix} \quad [K_{ph}] = \text{diag} \{ k_1 ; k_2 \}$$

so we express linearized relationship

$$\underline{y}_K = [\lambda_{k_0}] \underline{x}$$

↑
LINEARIZED @ @ EQUILIBRIUM

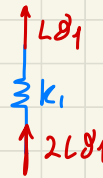
$$\lambda_{k_0} = \begin{bmatrix} \theta_1 & \theta_2 \\ -L & -L + 2R \\ -L & -R \end{bmatrix} \begin{matrix} \Delta l_1 \\ \Delta l_2 \end{matrix}$$

as previously

$$\bullet \theta_1 \neq 0, \theta_2 = 0$$

fixed disk all points of disk moves as G_1 of $L\theta_1$

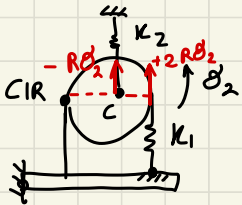
$$\text{while } \Delta l_1 = \underbrace{-2L\theta_1}_{\text{move of D}} + \underbrace{L\theta_1}_{\text{move of B}} = -L\theta_1$$



While k_2 shrunk of $-L\theta_1$



$$\bullet \theta_1 = 0, \theta_2 \neq 0$$



$$[K] = [\lambda_{k_0}]^T [K_{ph}] [\lambda_{k_0}] =$$

$$\begin{bmatrix} L^2(k_1 + k_2) & LR(-2k_1 + k_2) \\ LR(-2k_1 + k_2) & R^2(4k_1 + k_2) \end{bmatrix}$$

DISSIPATION.

as before we see that each damper is parallel to each spring so some

$$[C] = [\lambda_{k_0}]^T [C_{ph}] [\lambda_{k_0}] =$$

$$\begin{bmatrix} L^2(c_1 + c_2) & LR(-2c_1 + c_2) \\ LR(-2c_1 + c_2) & R^2(4c_1 + c_2) \end{bmatrix}$$

• VIRTUAL WORK (C Torque acts on the Disk) (NO NET FORCE)

system of FORCES with $\sum \vec{F} = 0$ BUT
with a total moment $\neq 0$

$$\delta W_{nc} = \delta \theta_2 \cdot C =$$

Im finit.
ROTATION
ON DISK
(same act
the TORQUE)

L₁
for
example



$$\sum f = 0$$

BUT moment
respect any center $\neq 0$
 $f \cdot a$ is the TORQUE

$$C = f \cdot a \quad (a: \text{ARM})$$

$$= \delta y_F^T f_{ph}$$

$$\delta y_F^T = (\delta \theta_2) \quad f_{ph} = (C)$$

and we can explicitly
write... JACOBIAN of this

VIRTUAL variation

$$\delta y_F = [\lambda_F] \cdot \delta x$$



$$[\lambda_F] = \begin{bmatrix} \delta \theta_1 & \delta \theta_2 \\ 0 & 1 \end{bmatrix} \delta \theta_2$$

easy TRIVIAL case,
JUST identity
($\theta_2 = \theta_2$)

$$\delta W_{nc} = \delta x^T \underbrace{f(t)}_{\text{RIGHT hand side of eq of motion}}$$

$$\underline{f}(t) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} C(t) = \begin{bmatrix} 0 \\ C(t) \end{bmatrix}$$

We found

all the MATRIX, part of our Linearized eq of motion
valid for small BAR ROTATION θ_1

if θ_1 BIG ROTATION NOT Good equations!

small θ_1
around $\theta_1 = 0$

$$[m] \ddot{x} + [c] \dot{x} + [k] x = \underline{f}(t)$$